CE205 Databases and Information Retrieval

Lab 1

(Week 3)

Database for Marcia's Dry Cleaning

Marcia owns and operates Marcia's Dry Cleaning, which is an upscale dry cleaner in a well-to-do suburban neighbourhood. Marcia makes her business stand out from the competition by providing superior customer service. She wants to keep track of each of her customers and their orders. Ultimately, she wants to notify them that their clothes are ready via e-mail. To provide this service, she has developed an initial database with several tables. Three of those tables are the following:

CUSTOMER (CustomerID, FirstName, LastName, Phone, Email)

INVOICE (InvoiceNumber, *CustomerNumber*, DateIn, DateOut, TotalAmount)

INVOICE_ITEM (*InvoiceNumber*, ItemNumber, Item, Quantity, UnitPrice)

In the above relations, the primary keys are underlined and the foreign keys are shown in italics. The database that Marcia has created is named MDC, and the three tables in the MDC database schema.

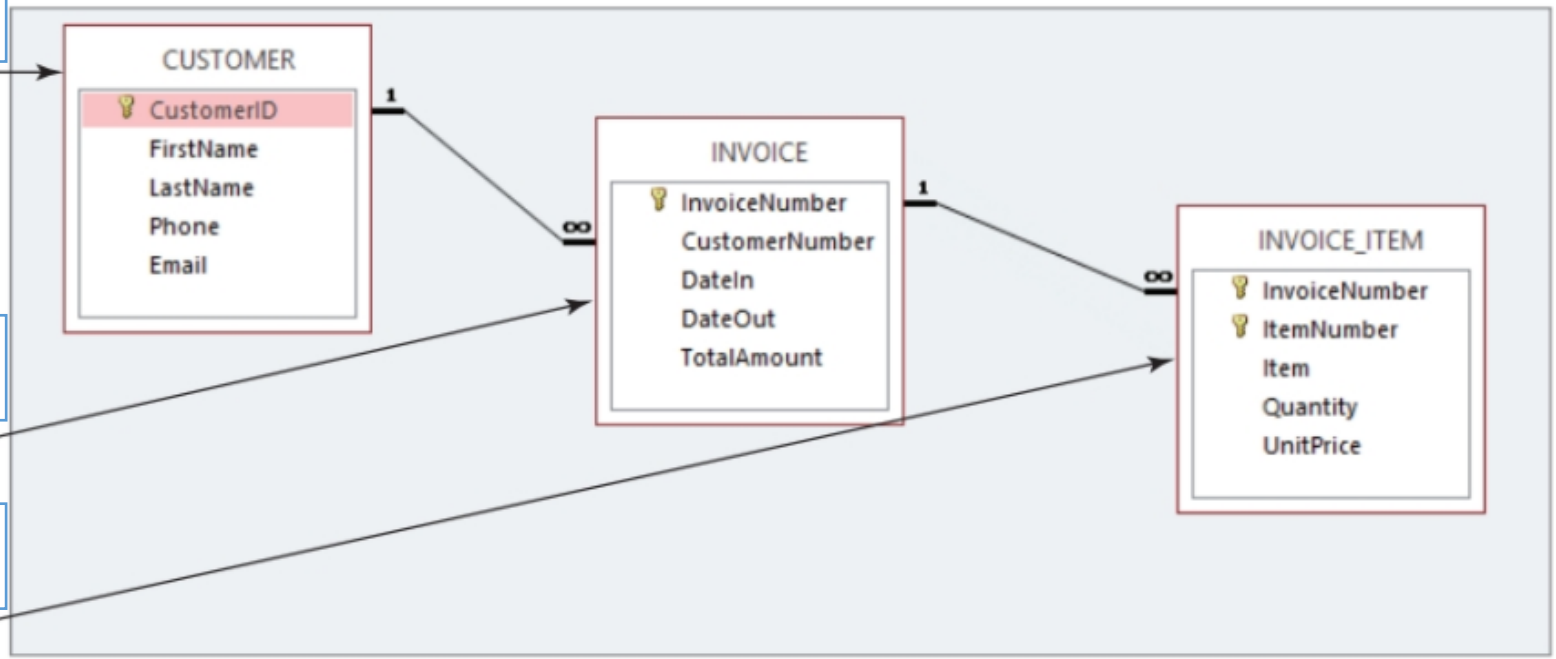
Taken from *DATABASE PROCESSING: Fundamentals*, Design, and Implementation, 15th Edition, by Kroenke, Auer, Vandenberg, Yoder

Relational Schema of MDC

The CUSTOMER table

The INVOICE table

The INVOICE_ITEM table



Column Characteristics of Tables in MDC

The column characteristics for the tables are given below. The relationship between CUSTOMER and INVOICE should enforce referential integrity, but not cascade updates nor deletions, while the relationship between INVOICE and INVOICE_ ITEM should enforce referential integrity and cascade both updates and deletions.

CUSTOMER

Column Name	Type	Key	Required	Remarks
CustomerID	Integer	Primary Key	Yes	Surrogate Key
FirstName	Character (25)	No	Yes	
LastName	Character (25)	No	Yes	
Phone	Character (12)	No	No	
Email	Character (100)	No	No	Use Varchar

INVOICE

Column Name	Type	Key	Required	Remarks
InvoiceNumber	Integer	Primary Key	Yes	Surrogate Key
CustomerNumber	Integer	Foreign Key	Yes	REF: CUSTOMER
DateIn	Date	No	Yes	
DateOut	Date	No	No	
TotalAmount	Number (8,2)	No	No	

INVOICE_ITEM

Column Name	Type	Key	Required	Remarks
InvoiceNumber	Integer	Primary Key, Foreign Key	Yes	REF: INVOICE
ItemNumber	Integer	Primary Key	Yes	Sequential number, but *not* a surrogate key
Item	Character (50)	No	Yes	
Quantity	Integer	No	Yes	
UnitPrice	Number (8,2)	No	Yes	

Sample Data for Tables in MDC (1)

CUSTOMER

CustomerID	FirstName	LastName	Phone	Email
1	Nikki	Kaccaton	723-543-1233	Nikki.Kaccaton@somewhere.com
2	Brenda	Catnazaro	723-543-2344	Brenda.Catnazaro@somewhere.com
3	Bruce	LeCat	723-543-3455	Bruce.LeCat@somewhere.com
4	Betsy	Miller	725-654-3211	Betsy.Miller@somewhere.com
5	George	Miller	725-654-4322	George.Miller@somewhere.com
6	Kathy	Miller	723-514-9877	Kathy.Miller@somewhere.com
7	Betsy	Miller	723-514-8766	Betsy.Miller@elsewhere.com

INVOICE

InvoiceNumber	CustomerNumber	DateIn	DateOut	TotalAmount
2015001	1	04-Oct-15	06-Oct-15	$158.50
2015002	2	04-Oct-15	06-Oct-15	$25.00
2015003	1	06-Oct-15	08-Oct-15	$49.00
2015004	4	06-Oct-15	08-Oct-15	$17.50
2015005	6	07-Oct-15	11-Oct-15	$12.00
2015006	3	11-Oct-15	13-Oct-15	$152.50
2015007	3	11-Oct-15	13-Oct-15	$7.00
2015008	7	12-Oct-15	14-Oct-15	$140.50
2015009	5	12-Oct-15	14-Oct-15	$27.00

Sample Data for Tables in MDC (2)

INVOICE_ITEM

InvoiceNumber	ItemNumber	Item	Quantity	UnitPrice
2015001	1	Blouse	2	$3.50
2015001	2	Dress Shirt	5	$2.50
2015001	3	Formal Gown	2	$10.00
2015001	4	Slacks-Mens	10	$5.00
2015001	5	Slacks-Womens	10	$6.00
2015001	6	Suit-Mens	1	$9.00
2015002	1	Dress Shirt	10	$2.50
2015003	1	Slacks-Mens	5	$5.00
2015003	2	Slacks-Womens	4	$6.00
2015004	1	Dress Shirt	7	$2.50
2015005	1	Blouse	2	$3.50
2015005	2	Dress Shirt	2	$2.50
2015006	1	Blouse	5	$3.50
2015006	2	Dress Shirt	10	$2.50
2015006	3	Slacks-Mens	10	$5.00
2015006	4	Slacks-Womens	10	$6.00
2015007	1	Blouse	2	$3.50
2015008	1	Blouse	3	$3.50
2015008	2	Dress Shirt	12	$2.50
2015008	3	Slacks-Mens	8	$5.00
2015008	4	Slacks-Womens	10	$6.00
2015009	1	Suit-Mens	3	$9.00

Once you have setup MDC database, create an SQL script named MDC.sql, and use it to record and store SQL statements that answer each of the following questions (if the question requires a written answer, use an SQL comment to record your answer).

Suggested Resources:

https://dev.mysql.com/doc/refman/5.7/en/sql-statements.html

https://www3.ntu.edu.sg/home/ehchua/programming/sql/MySQL_Beginner.html

Questions

A. Show all data in each of the tables.

B. List the LastName, FirstName, and Phone of all customers.

C. List the LastName, FirstName, and Phone for all customers with a FirstName of 'Nikki'.

D. List the LastName, FirstName, Phone, DateIn, and DateOut of all orders in excess of $100.00.

E. List the LastName, FirstName, and Phone of all customers whose first name starts with 'B'.

F. List the LastName, FirstName, and Phone of all customers whose last name includes the characters 'cat'.

G. List the LastName, FirstName, and Phone for all customers whose second and third digits (from the left) of their phone number are 23. For example, any phone number with an area code of "723" would meet the criteria.

H. Determine the maximum and minimum TotalAmount.

I. Determine the average TotalAmount.

J. Count the number of customers.

K. Group customers by LastName and then by FirstName.

L. Count the number of customers having each combination of LastName and FirstName.

M. Show the LastName, FirstName, and Phone of all customers who have had an order with TotalAmount greater than $100.00. Use a subquery. Present the results sorted by LastName in ascending order and then FirstName in descending order.

N. Show the LastName, FirstName, and Phone of all customers who have had an order with TotalAmount greater than $100.00. Use a join, but do not use JOIN ON syntax. Present results sorted by LastName in ascending order and then FirstName in descending order.

O. Show the LastName, FirstName, and Phone of all customers who have had an order with TotalAmount greater than $100.00. Use a join using JOIN ON syntax. Present results sorted by LastName in ascending order and then FirstName in descending order.

P. Show the LastName, FirstName, and Phone of all customers who have had an order with an Item named 'Dress Shirt'. Use a subquery. Present results sorted by LastName in ascending order and then FirstName in descending order.

Q. Show the LastName, FirstName, and Phone of all customers who have had an order with an Item named 'Dress Shirt'. Use a join, but do not use JOIN ON syntax. Present results sorted by LastName in ascending order and then FirstName in descending order.

R. Show the LastName, FirstName, and Phone of all customers who have had an order with an Item named 'Dress Shirt'. Use a join using JOIN ON syntax. Present results sorted by LastName in ascending order and then FirstName in descending ord sorted by LastName in ascending order and then FirstName in descending order.

S. Show the LastName, FirstName, and Phone of all customers who have had an order with an Item named 'Dress Shirt'. Use a combination of a join using JOIN ON syntax and a subquery. Present results sorted by LastName in ascending order and then FirstName in descending order.

T. Show the LastName, FirstName, Phone, and TotalAmount of all customer orders that included an Item named 'Dress Shirt'. Also show the LastName, FirstName, and Phone of all other customers. Present results sorted by TotalAmount in ascending order, then LastName in ascending order, and then FirstName in descending order.
